

Q. What will be the de-Broglie wavelength of an electron having kinetic energy $1.6 \times 10^{-13} \text{ J}$ [$m_e = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.63 \times 10^{-34} \text{ Js}$]

Q. Calculate the de-Broglie wavelength of an α -particle having kinetic energy 2 eV .

Q. Calculate the de-Broglie wavelength of a neutron of energy 28.8 eV [$m_n = 1.67 \times 10^{-27} \text{ kg}$, $h = 6.63 \times 10^{-34} \text{ Js}$]

Q. What will be the de-Broglie wavelength
-th of an electron having kinetic
energy 50 eV [$m_e = 9.1 \times 10^{-31} \text{ kg}$,

$$h = 6.63 \times 10^{-34} \text{ Js}]$$

Q. Find the de - Broglie wavelength associated with a bullet of mass m which is moving with a velocity of 50 m/s .

$$h = 6.63 \times 10^{-34} \text{ Js}$$

8. What is the de-Broglie wavelength of an electron that is accelerated from rest by a potential difference of 150 V?

Mass of $e^- = 9.1 \times 10^{-31} \text{ kg}$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

Q. Show that a given system, de-Broglie wavelength is inversely proportional to
 (1) the square root of its mass (2) the sq. root of its kinetic energy.

* Black Body radiation & Photoelectric effect.